Polytope Fitting of Aggregated Heterogeneous Flexibility Regions via Approximate Gradient Descent

Yilin Wen, *Member, IEEE,* Xinyao Yang, Bo Zhao, *Member, IEEE,* Zhe Chen, and Gengming Liu, *Member, IEEE*

***Abstract*—Aggregating heterogeneous flexibility resources in smart grids remains challenging due to the reliance of existing methods on predefined model templates or problem-specific algorithms, which limits their generality and approximation quality. This paper proposes a unified polytope fitting framework for aggregated flexibility regions, where both the geometry matrix and the boundary vector are optimized as decision variables to minimize fitting error. The fitting error is characterized by two complementary metrics, namely feasibility error and optimality error, both of which admit tractable approximate gradients, enabling an efficient approximate gradient descent algorithm. The proposed method consistently outperforms existing approaches in both feasibility and optimality metrics. Moreover, the proposed method scales well to large populations, supports mixed-integer resource models, and effectively handles uncertainty. Case studies across diverse scenarios validate the effectiveness of the proposed method.**

***Index Terms*—Flexibility aggregation, polytope fitting, heterogeneous resources, approximate gradient descent.**

I. Introduction

THE INCREASING penetration of volatile renewable energy sources poses significant challenges to the secure and efficient operation of modern power systems. In this context, effectively harnessing the flexibility of heterogeneous distributed resources has become a key enabler for balancing electricity supply and demand. However, individual flexible resources are typically characterized by limited controllable capacity, high diversity, and uncertainty, while also raising concerns related to user privacy and communication constraints. As a result, from the perspective of system operators, it is often impractical to precisely observe and directly control each individual device. Instead, system operators are primarily concerned with the aggregated operational region of multiple resources, in order to assess the total flexibility that aggregation entities, such as microgrids [1], load aggregators [2], and virtual power plants (VPPs) [3], can provide to the power system.

Extensive efforts have been devoted to the flexibility aggregation problem, which can be broadly classified into template-based and generative approaches. The template-based approach assumes a predefined model template and then determines its parameters accordingly. For example, [4] proposed a box-type flexibility region characterized by upper and lower power bounds, with parameters obtained via mathematical optimization. This template has also been adopted in [5]–[7], where the parameter estimation process is further enhanced using data-driven machine learning techniques. In addition, [8]–[10] employ storage templates to construct aggregated models, where different computational methods are developed to identify the unknown parameters within the assumed structures. Reference [11] introduced a power-energy boundary model to uniformly describe the operational feasibility of electric vehicles (EVs), thermostatically controlled loads (TCLs), energy storage systems (ESSs), and photovoltaics (PVs), and derived an outer approximation of the aggregated flexibility through parameter superposition. Under similar power-energy boundary assumptions, [12] derived an exact analytical expression for the aggregated model, but the resulting formulation exhibits exponential complexity, limiting its direct applicability. To reduce complexity, [13] proposed an iterative boundary refinement algorithm that converges to an inner approximation of the aggregated flexibility region. On the other hand, [14] extended the power-energy boundary framework to explicitly account for uncertainty. In [3] and [15], power-energy boundary models were adopted for storage-type resources, whereas power and ramp-rate boundary models were proposed for generation-type resources. In addition, [16] described aggregated active and reactive power flexibility using time-varying elliptical cylinders, whose parameters are computed via two-stage robust optimization. Several works have also focused specifically on a single type of resource, e.g., EVs [17], [18] and TCLs [19]–[21], and have proposed tailored models and computational methods. Moreover, zonotope models [22] and generalized operational models that simultaneously capture generation, storage, and thermal characteristics [23] have also been employed to model the aggregated flexibility regions of heterogeneous resources.

In summary, these works have developed a wide variety of effective modeling templates and algorithmic approaches for aggregating the flexibility of multiple resources. Most of these templates can be characterized by a set of predefined linear inequalities [3]–[15], [17]–[23], namely a polytopic set of the form

$$\mathcal{P}(\mathbf{A}, \mathbf{b}) = \{\mathbf{P} \mid \mathbf{A}\mathbf{P} \leq \mathbf{b}\}, \tag{1}$$

where $\mathbf{P} \in \mathbb{R}^T$ denotes the aggregated power profile over T time intervals, and $\mathbf{A} \in \mathbb{R}^{M \times T}$ and $\mathbf{b} \in \mathbb{R}^M$ specify the polytope parameters through M linear inequalities. Polytopic models are attractive due to their simple structure and strong expressive capability when a sufficient number of facets is used. However, in existing studies, the geometry matrix $\mathbf{A}$ is typically prescribed a priori based on empirical considerations or engineering intuition. In particular, $\mathbf{A}$ is often constructed

by mirroring the constraint structures of individual resource models, while the computation of $\mathbf{b}$ is tightly coupled with the specific choice of $\mathbf{A}$ or the underlying resource types. Nevertheless, such empirically designed $\mathbf{A}$ may fail to accurately capture the true aggregated feasible region, particularly in highly heterogeneous settings where defining a well-fitting template based solely on intuition becomes inherently challenging.

In contrast to the above approaches, the iterative cutting-plane generation method [24] and vertex generation method [25] do not rely on predefined polytope shapes. However, these methods only allow the addition of new cutting planes or vertices without revising previously generated ones, making their performance sensitive to the choice of initialization. Moreover, these two approaches have inherent limitations. The cutting-plane method [24] can only yield an outer approximation when the number of hyperplanes is limited, potentially leading to infeasible dispatch decisions. In contrast, the vertex-based method [25] requires an exponentially increasing number of vertices as the dimension grows, resulting in a substantial computational burden.

The proposed method still falls within the template-based framework. However, to overcome the limitations of existing studies, this work treats the polytope geometry matrix as an optimization variable. In this way, it identifies a polytope that better approximates the true aggregated flexibility region than conventional empirically designed templates. The main contributions of this work are threefold:

- To the best of our knowledge, this work is the first to formulate the polytope fitting problem for aggregating heterogeneous flexibility regions. The polytope parameters $(\mathbf{A}, \mathbf{b})$ are treated as decision variables to directly optimize its geometry, enabling the identification of a polytope that well fits the unknown true aggregated flexibility region.
- Two complementary error metrics, namely feasibility and optimality, are defined to quantify the fitting quality. Tractable approximations of both metrics are derived, making them suitable for optimization and providing the foundation for geometry-aware polytope design.
- An approximate gradient descent method is proposed to solve the polytope fitting problem, with its core being the derivation of explicit approximate gradients for both error metrics. Starting from empirically constructed polytopes, the algorithm iteratively updates the geometry matrix $\mathbf{A}$ and boundary vector $\mathbf{b}$, achieving significantly better approximations over existing template-based approaches in terms of both feasibility and optimality errors. The proposed method is sufficiently general to scale to large populations, supports mixed-integer resource models, and effectively handles uncertainty, making it well-suited for practical deployment.

Overall, the proposed method establishes a general, high-accuracy, and computationally efficient framework for representing complex aggregated flexibility regions of heterogeneous resources. Numerical tests in continuous resource models demonstrate that the proposed approach achieves significant accuracy improvements over existing methods, while tests in mixed-integer resource models highlight its generality.

Notations: We use bold letters (e.g., $\mathbf{x}$, $\mathbf{A}$) to denote vectors and matrices, and the corresponding regular letters with subscripts to denote their components. The operator $(\cdot)^\top$ denotes the transpose of a vector or matrix.

II. From Individual to Aggregated Flexibility Regions

This section first formulates the flexibility region aggregation problem as a Minkowski sum problem. Then, using two illustrative examples, we show that complex features in individual models, such as discrete variables and uncertainty, can be smoothed in the Minkowski sum operation. This observation motivates the use of polytopes for a tractable representation of the true aggregated flexibility region.

A. Definition of Individual and Aggregated Flexibility Regions

For each resource $i \in \mathcal{I}$, let $p_{i,t}$ denote its power at time $t \in [T]$, with the convention that $p_{i,t} > 0$ represents power generation (injection to the grid) and $p_{i,t} < 0$ represents power consumption (withdrawal from the grid). Let $\mathcal{F}_i \subseteq \mathbb{R}^T$ denote the individual flexibility region of resource i, defined as the set of all feasible power trajectories $\mathbf{p}_i = (p_{i,1}, \ldots, p_{i,T})$ that satisfy its physical operational constraints (e.g., governing physical equations, power and energy limits, and operational requirements). The specific constraint formulations for six representative resource types, including microturbines (MTs), PVs, ESSs, curtailable loads (CLs), transferable loads (TLs), and TCLs, are provided in the Appendix. These constraints ensure that $\mathcal{F}_i$ is a bounded set.

The aggregated flexibility region Ω is the set of all aggregated power profiles $\mathbf{P} = (P_1, P_2, \ldots, P_T) \in \mathbb{R}^T$ that can be realized by some combination of individual resource profiles $\mathbf{p}_i \in \mathcal{F}_i$ for all $i \in \mathcal{I}$. Formally, Ω can be described as the Minkowski sum of the individual regions:

$$\Omega = \left\{ \mathbf{P} \middle| P_t = \sum_{i \in \mathcal{I}} p_{i,t} \forall t \in [T], \quad \mathbf{p}_i \in \mathcal{F}_i, \forall i \in \mathcal{I} \right\}. \quad (2)$$

The above formulation is not a closed-form representation of Ω, as it still depends on the individual decision variables $\mathbf{p}_i$ and flexibility regions $\mathcal{F}_i$. Deriving a high-accuracy closed-form characterization of Ω, i.e., expressed solely in terms of the aggregate variable $\mathbf{P}$, constitutes the central objective of the aggregation problem.

B. Phenomena in Aggregating Heterogeneous Regions

The individual flexibility region $\mathcal{F}_i$ may exhibit intrinsic complexity due to two factors. First, $\mathcal{F}_i$ can be disconnected, arising from discrete operational variables such as on/off decisions for generators and charging/discharging modes of energy storage systems. Second, $\mathcal{F}_i$ may be subject to uncertainty, due to exogenous factors such as renewable generation, load variations, and ambient conditions. These characteristics make it intractable to derive a closed-form representation of the true aggregated region Ω. However, by examining the Minkowski

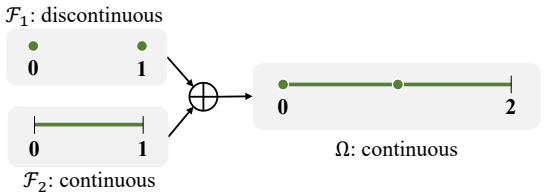


Fig. 1: Illustration of the Minkowski sum of a discontinuous set and a continuous set.

summation, we observe two key phenomena indicating that aggregation can mitigate these complexities. In particular, the resulting aggregated region becomes smoother, with more concentrated boundaries. These properties suggest that it is possible to construct a high-quality polytopic approximation of Ω.

1) The aggregation of mixed continuous-discontinuous regions may ultimately yield a continuous region.

This is because when the contribution of continuous regions is sufficiently dominant, their Minkowski sum can smooth out discrete gaps and yield a continuous aggregated region. Formally speaking, according to the Shapley–Folkman lemma [26], the Minkowski sum of sufficiently many sets will approximate a convex set. Therefore, the transition from discontinuity to continuity is a phenomenon resulting from this lemma.

To illustrate this effect, consider the example shown in Fig. 1. Let $\mathcal{F}_1 = \{0, 1\}$ be a discrete set consisting of two isolated points, and let $\mathcal{F}_2 = [0, 1]$ be a continuous interval. Their Minkowski sum yields the aggregated region $\Omega = [0, 2]$, which is a continuous interval. This example indicates that the binary decision variables at the individual-resource level, which are computationally challenging, may be effectively mitigated through aggregation, resulting in an aggregated region that is less complex than initially anticipated.

2) Conservative estimations of uncertain regions tend to be more reliable when constructed at the aggregation level rather than at the individual level.

In the presence of uncertainty, conservative estimation is commonly adopted to ensure feasibility. Due to statistical aggregation effects, the boundary distribution of the aggregated region is typically more concentrated than that of individual regions. As a result, applying the same level of conservatism directly to the aggregated region leads to a significantly lower probability of over-estimation compared to independently estimating each individual region.

This can be demonstrated by the example in Fig. 2. Consider the Minkowski sum of two uncertain regions $\mathcal{F}_1$ and $\mathcal{F}_2$, each given by the interval $[0, \tilde{u}_i](i \in \{1, 2\})$, where $\tilde{u}_i(i \in \{1, 2\})$ follows a uniform distribution on $[0.8, 1.2]$. If the same conservative estimate $\hat{u} = 0.9$ is applied to each individual region, the resulting aggregated estimate is $[0, 1.8]$, under which the probability that at least one realized individual region is not covered equals $\frac{1}{4} \cdot \frac{3}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{7}{16}$. In contrast, when modeling the aggregated region directly, its upper bound $\tilde{t} = \tilde{u}_1 + \tilde{u}_2$ follows a triangular distribution on $[1.6, 2.4]$. Using the same estimate interval $[0, 1.8]$, the probability that the true aggregated region is not covered reduces to $\frac{1}{8}$. This comparison highlights the advantage of addressing uncertainty at the aggregation level.

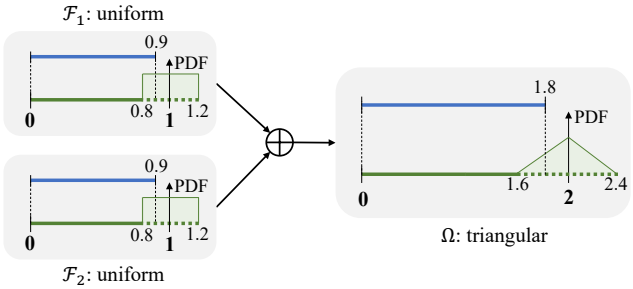


Fig. 2: Illustration of conservative estimation for uncertain regions at individual and aggregation levels. The blue line indicates the estimated intervals. The green dashed line represents the uncertain upper bounds, whose probability density function (PDF) is shown as the green shaded area.

These analyses jointly suggest that the aggregated region Ω, despite being formed from heterogeneous and possibly discontinuous and uncertain individual sets, often exhibits a smoothed, approximately convex, and concentrated geometric structure. Such properties provide a sound basis for the polytopic approximation of Ω, making it possible to expect high accuracy and reliability.

In the following discussion, we temporarily disregard uncertainty factors and assume that all uncertain variables take their expected values. We will extend the proposed method in Subsection IV-F to handle uncertainty.

III. Polytope Fitting Problem for Aggregating Flexibility Regions

A. Problem Definition

We seek to fit Ω using the polytopic region $\mathcal{P}(\mathbf{A}, \mathbf{b})$ in (1), where $\mathbf{A} \in \mathbb{R}^{M \times T}$ and $\mathbf{b} \in \mathbb{R}^{M}$ define M hyperplane facets of the polytope, and M is a design parameter chosen by the user (higher M offers more shape adaptability at the cost of greater complexity). The fitting task is to identify a pair $(\mathbf{A}, \mathbf{b})$ such that the polytope $\mathcal{P}(\mathbf{A}, \mathbf{b})$ closely approximates the true aggregated region Ω, with the discrepancy between them minimized.

To quantify the discrepancy between a candidate polytope $\mathcal{P}(\mathbf{A}, \mathbf{b})$ and the true region Ω, we define two complementary error metrics, i.e., the feasibility error and the optimality error. First, the feasibility error e_{feas} measures how much $\mathcal{P}(\mathbf{A}, \mathbf{b})$ extends outside the true region:

$$e_{\text{feas}}(\mathbf{A}, \mathbf{b}) \triangleq \max_{\mathbf{P}_{\text{apx}} \in \mathcal{P}(\mathbf{A}, \mathbf{b})} \min_{\mathbf{P}_{\text{org}} \in \Omega} \|\mathbf{P}_{\text{apx}} - \mathbf{P}_{\text{org}}\|^2, \quad (3)$$

which is the squared distance from the "furthest outside" polytope point $\mathbf{P}_{\text{apx}}$ to the nearest point in Ω. If $\mathcal{P}(\mathbf{A}, \mathbf{b})$ is entirely inside Ω, then $e_{\text{feas}} = 0$ (every power profile allowed by the polytopic region is feasible for the true aggregated flexibility region). Second, the optimality error e_{opt} measures how much of Ω lies outside the polytope:

$$e_{\text{opt}}(\mathbf{A}, \mathbf{b}) \triangleq \max_{\mathbf{P}_{\text{org}} \in \Omega} \min_{\mathbf{P}_{\text{apx}} \in \mathcal{P}(\mathbf{A}, \mathbf{b})} \|\mathbf{P}_{\text{org}} - \mathbf{P}_{\text{apx}}\|^2, \quad (4)$$

which is the squared distance from the "furthest missing" point $\mathbf{P}_{\text{org}} \in \Omega$ to its nearest polytope point $\mathbf{P}_{\text{apx}}$. If $e_{\text{opt}} = 0$, the polytope $\mathcal{P}(\mathbf{A}, \mathbf{b})$ covers the entire region Ω (i.e., for every feasible aggregate profile, the polytope contains an equal profile). In essence, $e_{\text{feas}} = 0$ ensures feasibility (the polytope is an inner approximation of Ω) and $e_{\text{opt}} = 0$ ensures optimality (the polytope is an outer approximation covering Ω). These two errors together relate to the Hausdorff distance [27] between $\mathcal{P}(\mathbf{A}, \mathbf{b})$ and Ω (i.e., the maximum of $\sqrt{e_{\text{feas}}}$ and $\sqrt{e_{\text{opt}}}$). However, directly minimizing the Hausdorff distance is not employed to identify the optimal polytope in the flexibility aggregation task. This is due firstly to the non-differentiability of the maximum operation, which compromises computational stability; and secondly to the fact that feasibility is prioritized over optimality in flexibility aggregation, precluding the symmetric treatment inherent in the Hausdorff distance.

Therefore, we choose to minimize a weighted sum of the feasibility and optimality errors as formulated in (5), which allows us to balance the trade-off between conservative and comprehensive fitting of Ω.

$$(\mathbf{A}^*, \mathbf{b}^*) = \arg\min_{\mathbf{A},\mathbf{b}} \lambda e_{\text{feas}}(\mathbf{A}, \mathbf{b}) + (1-\lambda) e_{\text{opt}}(\mathbf{A}, \mathbf{b}) \quad (5)$$

where $\lambda \in [0, 1]$ is a weighting parameter. By adjusting λ, we can emphasize different goals: setting $\lambda = 1$ focuses purely on feasibility (seeking to find a good inner approximation that sacrifices the maximum coverage of Ω), whereas $\lambda = 0$ would prioritize covering Ω entirely (seeking to find an outer approximation at the cost of infeasible aggregated power profiles). An intermediate $0 < \lambda < 1$ strikes a balance, fitting the shape of Ω as closely as possible without overly violating either criterion. In fact, λ can be varied during the optimization to first find a good overall fit and then emphasize feasibility (see the optimization algorithm in Subsection IV-B).

B. Tractable Reformulation based on Directional Errors

Solving problem (5) is challenging because evaluating $e_{\text{feas}}(\mathbf{A}, \mathbf{b})$ and $e_{\text{opt}}(\mathbf{A}, \mathbf{b})$ for a given $(\mathbf{A}, \mathbf{b})$ is generally NP-hard [27]. Therefore, a tractable reformulation is required to enable efficient computation. To this end, we introduce *directional errors*. The key idea is to decompose the problem along one-dimensional directions. Given a unit direction vector $\mathbf{v} \in \mathbb{R}^T$, the directional feasibility and optimality errors are defined as follows:

$$e_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v}) \triangleq \min_{\mathbf{P}_{\text{org}} \in \Omega} \|\mathbf{P}_{\text{org}} - \mathbf{P}'_{\text{apx}}\|^2, \quad (6a)$$

$$\text{s.t.} \quad \mathbf{P}'_{\text{apx}} = \arg\min_{\mathbf{P}_{\text{apx}} \in \mathcal{P}(\mathbf{A},\mathbf{b})} \mathbf{v}^\top \mathbf{P}_{\text{apx}}, \quad (6b)$$

$$e_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v}) \triangleq \min_{\mathbf{P}_{\text{apx}} \in \mathcal{P}(\mathbf{A},\mathbf{b})} \|\mathbf{P}_{\text{apx}} - \mathbf{P}'_{\text{org}}\|^2, \quad (7a)$$

$$\text{s.t.} \quad \mathbf{P}'_{\text{org}} = \arg\min_{\mathbf{P}_{\text{org}} \in \Omega} \mathbf{v}^\top \mathbf{P}_{\text{org}}. \quad (7b)$$

By definition, both $e_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ and $e_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ can be calculated by solving two optimization problems separately. In particular, to obtain $e_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$, we first find the polytope point $\mathbf{P}'_{\text{apx}}$ that is extremal in direction $\mathbf{v}$ by solving (6b), and then measure the squared distance from $\mathbf{P}'_{\text{apx}}$ to the true region Ω by solving (6a). Similarly, $e_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ is obtained by solving (7b) first and then (7a). Each computation involves solving explicit optimization problems over the linear polytope $\mathcal{P}(\mathbf{A}, \mathbf{b})$ or the original region Ω with linear or quadratic objectives, which can be handled efficiently with standard solvers.

The directional errors are closely related to the original feasibility and optimality errors defined in (3) and (4). Let $\mathbb{S}^{T-1} = \{\mathbf{v} \in \mathbb{R}^T : \|\mathbf{v}\| = 1\}$ denote the unit sphere in $\mathbb{R}^T$. Assuming that the aggregated flexibility region Ω is convex (which is reasonable according to the analysis in Subsection II-B, even if the individual regions are nonconvex or disconnected), the original error metrics can be recovered by taking the maximum of the directional errors over all directions:

$$e_{\text{feas}}(\mathbf{A}, \mathbf{b}) = \max_{\mathbf{v} \in \mathbb{S}^{T-1}} e_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v}), \quad (8)$$

$$e_{\text{opt}}(\mathbf{A}, \mathbf{b}) = \max_{\mathbf{v} \in \mathbb{S}^{T-1}} e_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v}). \quad (9)$$

This equivalence holds because, for convex sets, the support function fully characterizes the set geometry, and the extremal points in each direction capture the boundary information. However, when Ω is nonconvex, the relationship no longer holds directly, as the directional optimality error in (7) only measures the discrepancy between $\mathcal{P}(\mathbf{A}, \mathbf{b})$ and the convex hull of Ω. As a result, the directional error formulation may guide the polytope to approximate the convex hull of the true aggregated region. This is not a significant limitation in practice, since the convex hull is often sufficient for power system applications, and the associated approximation error can be further reduced by aggregating a larger number of flexible resources.

It is worth noting that (8) and (9) introduce an additional maximization over directions on top of (6) and (7), respectively, which again makes it intractable to directly solve for $e_{\text{feas}}(\mathbf{A}, \mathbf{b})$ and $e_{\text{opt}}(\mathbf{A}, \mathbf{b})$. Fortunately, solving (8) and (9) is not necessary in practice, as directional errors themselves already provide sufficient information for optimizing the polytope. In particular, since the polytope may deviate from the true region differently across directions, any identified discrepancy along a sampled direction can be used to update $\mathbf{A}$ and $\mathbf{b}$. Rather than searching for the global maximum over the entire unit sphere $\mathbb{S}^{T-1}$, we adopt a more efficient strategy that samples directions and iteratively refines the polytope based on the observed mismatches. This approach, i.e., leveraging local mismatches rather than pursuing the global worst-case direction, forms the basis of the iterative optimization algorithm presented in the next section.

IV. Optimization via Approximate Gradient Descent

While the directional errors tell us how well the polytope $\mathcal{P}(\mathbf{A}, \mathbf{b})$ fits Ω in each direction, we also need a way to update the polytope parameters $(\mathbf{A}, \mathbf{b})$ to improve the fit. A straightforward approach would be to employ gradient-based optimization. However, the directional errors depend on $\mathbf{A}$

and $\mathbf{b}$ only implicitly through two optimization problems in (6) and (7), which makes it difficult to obtain numerically stable gradients. To this end, this section develops an efficient solution algorithm based on approximate gradient descent.

The following subsections first introduce the proposed approximate error function and its corresponding gradients, then present the complete optimization procedure. We also provide a convergence analysis of the proposed method, and finally discuss how uncertainty is incorporated into the framework.

A. Approximate Error Function and Gradients

The approximate error function is constructed from the boundary points identified during the directional error calculation process. The key idea is to exploit the local active constraints of the polytope at these boundary points. We explain this construction using the feasibility error as an example.

For a given direction $\mathbf{v}$, the polytope boundary point solved by (6b) is $\mathbf{P}'_{\text{apx}}$. Let J be the index set of hyperplane constraints of the polytope $\mathcal{P}(\mathbf{A}, \mathbf{b})$ that are binding at $\mathbf{P}'_{\text{apx}}$, i.e., $\mathbf{A}_J\mathbf{P}'_{\text{apx}} = \mathbf{b}_J$. Let $\mathbf{P}^\star_{\text{org}}$ denote the nearest point in Ω solved by (6a). Assuming that each row vector $\mathbf{A}_j, \forall j \in J$ is normalized to unit length, we define the approximate directional feasibility error as

$$L_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v}) \triangleq \left\|\mathbf{A}_J\mathbf{P}^\star_{\text{org}} - \mathbf{b}_J\right\|^2. \tag{10}$$

This quantity measures the deviation of $\mathbf{P}^\star_{\text{org}}$ from the active supporting hyperplanes at $\mathbf{P}'_{\text{apx}}$ through a quadratic function with respect to the selected rows of $\mathbf{A}$ and $\mathbf{b}$. It therefore provides a smooth proxy for the directional feasibility error.

Using basic spatial geometry, the relationship between the approximate error function $L_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ and the original directional feasibility error $e_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ can be expressed as

$$L_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v}) = e_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v}) \sum\nolimits_{j\in J} \sin^2\phi_j, \tag{11}$$

where ϕ_j denotes the angle between the direction vector $\mathbf{P}^\star_{\text{org}} - \mathbf{P}'_{\text{apx}}$ and the hyperplane defined by $\mathbf{A}_j\mathbf{P} = b_j$.

Equation (11) shows that the approximate error L_{feas} differs from the original directional feasibility error e_{feas} only by a multiplicative factor determined by the local geometry of the active hyperplane. It is worth noting that the factor $\sum_{j\in J}\sin^2\phi_j = 0$ holds only when the displacement vector $\mathbf{P}^\star_{\text{org}} - \mathbf{P}'_{\text{apx}}$ is parallel to all active hyperplanes $\mathbf{A}_J\mathbf{P} = \mathbf{b}_J$. Since $\mathbf{A}_J\mathbf{P}'_{\text{apx}} = \mathbf{b}_J$, this parallelism further implies $\mathbf{A}_J\mathbf{P}^\star_{\text{org}} = \mathbf{b}_J$, i.e., $\mathbf{P}^\star_{\text{org}}$ also lies in the affine subspace defined by $\mathbf{A}_J\mathbf{P} = \mathbf{b}_J$. This corresponds to a highly degenerate configuration. Under generic conditions on the sample points, direction vectors, and polytope parameters, such exact alignment occurs with probability zero. Therefore, in almost all practical cases, $\sum_{j\in J}\sin^2\phi_j > 0$, implying that L_{feas} is almost surely positively correlated with e_{feas}, which justifies its use as a valid surrogate.

For the optimality error, let $\mathbf{P}'_{\text{org}}$ denote the extreme point of Ω in direction $\mathbf{v}$, and $\mathbf{P}^\star_{\text{apx}}$ the closest point on the polytope (with active facet set K). We define

$$L_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v}) \triangleq \|\mathbf{A}_K\mathbf{P}'_{\text{org}} - \mathbf{b}_K\|^2, \tag{12}$$

which provides a smooth measure of the extent to which $\mathbf{P}'_{\text{org}}$ violates the polytope constraints.

The two error terms are combined into a full approximate error function:

$$\begin{aligned} L(\mathbf{A}, \mathbf{b}, \mathbf{v}) &\triangleq \lambda L_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v}) + (1-\lambda)L_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v}) \\ &= \lambda\|\mathbf{A}_J\mathbf{P}^\star_{\text{org}} - \mathbf{b}_J\|^2 + (1-\lambda)\|\mathbf{A}_K\mathbf{P}'_{\text{org}} - \mathbf{b}_K\|^2. \end{aligned} \tag{13}$$

In principle, $\mathbf{P}'_{\text{org}}$, $\mathbf{P}^\star_{\text{org}}$, as well as the index sets J and K, depend implicitly on $\mathbf{A}$ and $\mathbf{b}$ through (6) and (7). However, within each iteration, these quantities are treated as fixed to facilitate gradient computation. This treatment corresponds to a straight-through approximation, which is commonly adopted to bypass non-differentiable or implicit dependencies during backpropagation [28]. The underlying rationale is detailed as follows.

First, $\mathbf{P}'_{\text{org}}$ and $\mathbf{P}^\star_{\text{org}}$ are boundary points of the true aggregated region Ω. Under small local perturbations of $\mathbf{A}$ and $\mathbf{b}$, their exact locations may change, but such changes typically correspond to continuous movements within a local neighborhood of the boundary, or occasional switches between adjacent facets. Under infinitesimal perturbations of $\mathbf{A}$ and $\mathbf{b}$, it is not necessary for these points to exactly coincide with the updated polytope. Instead, it suffices that they provide consistent directional information that drives the polytope boundary toward the true aggregated region Ω. Therefore, even when treated as fixed, these points still capture valid boundary deviation and lead to effective updates. Second, fixing the active constraint sets J and K admits a more direct justification. For linear inequality systems, results from parametric programming and the local stability of KKT conditions imply that the active set remains unchanged within a neighborhood of $(\mathbf{A}, \mathbf{b})$, since inactive constraints are separated from the active ones by a nonzero margin.

Under these approximations, explicit expressions for the approximate gradients $\nabla_{\mathbf{A}}L$ and $\nabla_{\mathbf{b}}L$ can be derived:

$$\nabla_{\mathbf{A}}L = \lambda\nabla_{\mathbf{A}}L_{\text{feas}} + (1-\lambda)\nabla_{\mathbf{A}}L_{\text{opt}}, \tag{14}$$

$$\nabla_{\mathbf{b}}L = \lambda\nabla_{\mathbf{b}}L_{\text{feas}} + (1-\lambda)\nabla_{\mathbf{b}}L_{\text{opt}}, \tag{15}$$

where

$$\nabla_{\mathbf{A}_j}L_{\text{feas}} \approx \begin{cases} 2(\mathbf{A}_j^\top\mathbf{P}^\star_{\text{org}} - b_j)\mathbf{P}^{\star\top}_{\text{org}}, j \in J \\ \mathbf{0}^\top \quad \text{, otherwise} \end{cases}, \tag{16}$$

$$\nabla_{b_j}L_{\text{feas}} \approx \begin{cases} -2(\mathbf{A}_j^\top\mathbf{P}^\star_{\text{org}} - b_j), j \in J \\ 0 \quad \text{, otherwise} \end{cases}, \tag{17}$$

$$\nabla_{\mathbf{A}_k}L_{\text{opt}} \approx \begin{cases} 2(\mathbf{A}_k^\top\mathbf{P}'_{\text{org}} - b_k)\mathbf{P}'^\top_{\text{org}}, k \in K \\ \mathbf{0}^\top \quad \text{, otherwise} \end{cases}, \tag{18}$$

$$\nabla_{b_k}L_{\text{opt}} \approx \begin{cases} -2(\mathbf{A}_k^\top\mathbf{P}'_{\text{org}} - b_k), k \in K \\ 0 \quad \text{, otherwise} \end{cases}. \tag{19}$$

From an implementation perspective, in addition to directly evaluating the gradients using (14)-(19), we can also employ automatic differentiation tools such as PyTorch [29] or TensorFlow [30] to simplify the implementation. Although these general tools are slightly less efficient than the explicit formulas, they offer greater convenience in terms of coding effort.

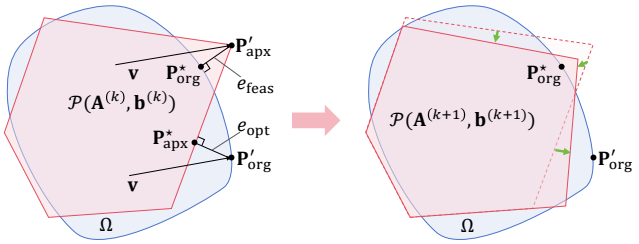


Fig. 3: Illustration of polytope modification after one iteration. Green arrows in the right subfigure indicate modifications to the polytope edges.

B. Optimization Algorithm

With the approximate error function and its gradients, we can now outline the basic optimization algorithm to solve problem (5) as follows:

1) **Initialization:** Set a maximum number of iterations $K_{\max}$ and a weight $\lambda \in (0, 1)$. Choose an initial polytope $\mathcal{P}(\mathbf{A}^{(0)}, \mathbf{b}^{(0)})$ that is a rough approximation of Ω.
2) **Iteration:** At each iteration $k = 0, 1, \ldots, K_{\max}$:
3) **Sample a direction:** Randomly select a direction $\mathbf{v}^{(k)}$ in $\mathbb{R}^T$ from a standard normal distribution $\mathcal{N}(0, \mathbf{I})$ in the T-dimensional space.
4) **Compute directional errors:** Compute the directional errors $e_{\text{feas}}(\mathbf{A}^{(k)}, \mathbf{b}^{(k)}, \mathbf{v}^{(k)})$ and $e_{\text{opt}}(\mathbf{A}^{(k)}, \mathbf{b}^{(k)}, \mathbf{v}^{(k)})$ according to (6) and (7).
5) **Compute approximate error and gradients:** Calculate the approximate error $L(\mathbf{A}^{(k)}, \mathbf{b}^{(k)}, \mathbf{v}^{(k)})$ using (13) and its gradients $\nabla_{\mathbf{A}^{(k)}} L$, $\nabla_{\mathbf{b}^{(k)}} L$ using Eqs. (14)-(19) or automatic differentiation tools.
6) **Update polytope parameters:** Update the parameters $(\mathbf{A}^{(k+1)}, \mathbf{b}^{(k+1)})$ using gradient-based optimizers, e.g., stochastic gradient descent (SGD) [31] or Adam [32]. Normalize the updated $(\mathbf{A}^{(k+1)}, \mathbf{b}^{(k+1)})$. Return to Step 2 until reaching the maximum iteration $K_{\max}$.

Fig. 3 illustrates the improvement of the polytope region through a single iteration (represented as a polygon in the 2D diagram). In each iteration, the identified point drives the adjustment of multiple edges of the polygon in directions that minimize the feasibility and optimality errors, thereby achieving a better fit to the true aggregated flexibility region. Since the sampled direction $\mathbf{v}^{(k)}$ varies across iterations, the active sets J and K change accordingly. Over many iterations, the optimization process provides opportunities for all facets to be refined, thereby improving the approximation of the original feasible region Ω from multiple directions.

The computational complexity of the proposed algorithm is primarily determined by the need to solve four optimization problems, namely (6a), (6b), (7a), and (7b), at each iteration. These correspond to linear and quadratic programs over the original region Ω and the approximating polytope $\mathcal{P}(\mathbf{A}, \mathbf{b})$, and do not introduce additional sources of complexity beyond standard optimization routines. Importantly, the framework does not rely on any structural assumptions about the original region. Even when the region is nonconvex or involves mixed-integer variables, the algorithm remains executable as long as appropriate nonlinear or mixed-integer solvers are available. In contrast, many existing approaches depend on analytical transformations of Ω [3], [4], [12], [24], which are typically not applicable in the presence of nonconvexities or integer variables. This highlights the generality of the proposed method.

C. Initialization

The optimization algorithm requires an initial polytope $\mathcal{P}(\mathbf{A}^{(0)}, \mathbf{b}^{(0)})$. In the literature, several well-established intuitive polytope models are available, such as the power boundary (PB) model [4]–[7] and the power-energy boundary (PEB) model [3], [11]–[15], both of which can serve as suitable initializations. We recommend the PEB model for initialization as it captures temporal coupling with low computational complexity ($M = 4T$). Specifically, under the PEB initialization, the matrix $\mathbf{A}$ is set as $\mathbf{A}^{(0)} = [\mathbf{I}, -\mathbf{I}, \mathbf{L}^\top, -\mathbf{L}^\top]^\top$, where $\mathbf{I}$ denotes the identity matrix, and $\mathbf{L}$ is constructed by normalizing each row of a lower-triangular matrix of ones to unit length. The vector $\mathbf{b}$ is then initialized by solving

$$b_j^{(0)} = \max_{\mathbf{P}_{\text{org}} \in \Omega} \mathbf{A}_j^{(0)} \mathbf{P}_{\text{org}}, \quad \forall 1 \le j \le M. \tag{20}$$

This initialization yields an outer approximation of Ω, i.e., $e_{\text{opt}}(\mathbf{A}^{(0)}, \mathbf{b}^{(0)}) = 0$.

If a smaller number of facets is desired (e.g., $M = 2T$), one may instead adopt the PB model as in [4], i.e., $\mathbf{A}^{(0)} = [\mathbf{I}, -\mathbf{I}]^\top$. In this case, when individual resource models are convex, the initialization corresponds to an inscribed hyperrectangle, which provides a feasible starting point for subsequent iterations. More generally, other polytope templates, such as those proposed in [12], can also be employed for initialization, provided that a sufficiently large number of facets M is allowed to achieve higher accuracy.

D. Discussion on Convergence

First, it should be emphasized that the polytope fitting problem is inherently highly nonconvex, due to the composite mapping: $(\mathbf{A}, \mathbf{b}) \to$ polytope $\to$ supporting points $\to$ projection $\to$ error metrics. Such nonconvexity is common in deep learning tasks, where no guarantee of convergence to a global optimum can be expected. Nevertheless, the proposed algorithm follows a stochastic gradient descent framework, which, under standard conditions, ensures convergence to a stationary point and thus provides a sound convergence basis.

This subsection mainly discusses the impact of the two key approximations on the convergence behavior of the algorithm:

1) using $L_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ and $L_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ to approximate the directional errors $e_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ and $e_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$;
2) neglecting the implicit dependence of $\mathbf{P}'_{\text{apx}}$ and $\mathbf{P}'_{\text{org}}$ on $\mathbf{A}$ and $\mathbf{b}$ in gradient computation.

For the first approximation, $L_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ and $L_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ are almost surely positively correlated with the directional errors $e_{\text{feas}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$ and $e_{\text{opt}}(\mathbf{A}, \mathbf{b}, \mathbf{v})$, respectively, and share the same minimizers, as previously discussed. Hence, minimizing the weighted sum of these approximations is consistent

with minimizing that of the directional errors, which preserves the correctness of the descent direction.

For the second approximation, neglecting the implicit dependence of $\mathbf{P}^{\star}_{\text{org}}$ and $\mathbf{P}'_{\text{org}}$ on $\mathbf{A}$ and $\mathbf{b}$ results in biased gradients. This type of approximation is common in engineering and machine learning. For example, in power systems, the fast decoupled power flow method approximates the Jacobian matrix to improve computational efficiency [33], while in machine learning, approximate or truncated gradients are commonly adopted to ensure tractability [28]. In the present context, each iteration effectively updates $(\mathbf{A}_J, \mathbf{b}_J)$ and $(\mathbf{A}_K, \mathbf{b}_K)$ to move the polytope boundary closer to the fixed reference points $\mathbf{P}^{\star}_{\text{org}}$ and $\mathbf{P}'_{\text{org}}$, respectively. This mechanism reduces the discrepancy between the active facets of the polytope and the true feasible boundary. Mathematically speaking, as $L(\mathbf{A}, \mathbf{b}, \mathbf{v})$ is lower bounded by zero, and decreases monotonically along the approximate gradient direction when the reference points are treated as fixed, under standard stochastic gradient descent assumptions, the expected approximate error $\mathbb{E}_{\mathbf{v}\sim\mathcal{N}(0,\mathbf{I})}[L(\mathbf{A}^{(k)}, \mathbf{b}^{(k)}, \mathbf{v})]$ converges as $k \to \infty$.

Over multiple iterations, different facets of the polytope are progressively refined, resulting in an increasingly accurate approximation of the feasible region. Although global optimality is not guaranteed, the obtained solution is sufficient for the polytope fitting task, as it consistently improves the polytope fit compared with the initialization based on state-of-the-art methods. This will be empirically validated in Section V, where numerical results demonstrate that the proposed method achieves significant accuracy improvements over existing methods.

E. Dynamic Weighting Strategy

Building upon this algorithm, we further incorporate a dynamic weighting strategy to drive the polytope toward a high-quality inner approximation. Specifically, we initially set $\lambda = 0.5$ to allow the feasibility and optimality errors to compete. This competition encourages the polytope to expand and contract in different directions to better fit Ω. Subsequently, λ is increased to larger values such as 0.9 and 0.9999 to emphasize feasibility.

As a result, the final polytope $\mathcal{P}(\mathbf{A}^*, \mathbf{b}^*)$ is, in practice, an inner approximation of the aggregated flexibility region. If the training process explores a sufficiently large number of directions $\mathbf{v}$, this approximation is essentially valid. Of course, from a theoretical standpoint, feasibility risks may still arise when only a limited number of training iterations are performed. However, our numerical experiments will demonstrate that the resulting feasibility errors are negligible, confirming that the proposed method is practically reliable. On the other hand, the polytope remains as "tight" as possible to minimize conservatism, thanks to the earlier shaping phases.

F. Data-Driven Solution for Uncertainty Handling

The individual flexibility region $\mathcal{F}_i$ of each flexible resource may be affected by various sources of uncertainty, e.g., PV generation, ambient temperature, and user behavior. These uncertainties lead to variations in the aggregated flexibility region Ω under different realizations. Let $\tilde{\boldsymbol{\xi}}$ denote the vector of uncertain parameters, then Ω is no longer a fixed region but rather a family $\Omega(\tilde{\boldsymbol{\xi}})$ of possible regions corresponding to different uncertainty realizations. For reliable power system operation, it is desirable that the aggregated flexibility model be robust against such uncertainties.

If a historical dataset $\boldsymbol{\Xi}$ or a probability distribution for $\tilde{\boldsymbol{\xi}}$ is available (a realistic assumption given the availability of historical data), the optimization algorithm in Subsection IV-B can be straightforwardly extended in a data-driven manner to handle uncertainty. Specifically, we introduce an additional step before Step 3: at each iteration k, draw a random sample $\boldsymbol{\xi}^{(k)}$ from the dataset $\boldsymbol{\Xi}$ or the distribution of $\tilde{\boldsymbol{\xi}}$. We then use the corresponding region $\Omega(\boldsymbol{\xi}^{(k)})$ to compute the directional errors $e_{\text{feas}}(\mathbf{A}^{(k)}, \mathbf{b}^{(k)}, \mathbf{v}^{(k)})$ and $e_{\text{opt}}(\mathbf{A}^{(k)}, \mathbf{b}^{(k)}, \mathbf{v}^{(k)})$ for that iteration (in place of using a fixed Ω). The rest of the algorithm (gradient computation and polytope update) proceeds as usual.

Over many iterations, the algorithm encounters a wide range of uncertainty realizations. As a result, the final solution polytope, in expectation, minimizes the weighted error metric over the underlying uncertainty distribution. Following the dynamic weighting strategy, λ is gradually increased to a value close to 1 in the later phases of training to prioritize feasibility. Consequently, the resulting polytope $\mathcal{P}(\mathbf{A}^*, \mathbf{b}^*)$ minimizes feasibility errors across diverse uncertainty realizations, thereby achieving strong robustness. After such training, at least over the training dataset, any aggregate profile $\mathbf{P} \in \mathcal{P}(\mathbf{A}^*, \mathbf{b}^*)$ can be decomposed with small feasibility error under various uncertainty realizations. However, when evaluated on unseen test data, infeasibility may still occur. A simple and practical method is to introduce conservatism to the individual flexibility models. For example, one may shrink the feasible ranges of PVs, CLs, or temperature control intervals of TCLs compared to their nominal limits. In this way, even if the uncertainty realizations deviate from the training distribution, the aggregated polytope remains sufficiently robust to maintain feasibility.

For extremely high levels of uncertainty, a strictly robust inner approximation may not exist (i.e., the inner approximation could be empty). In such cases, one can instead choose a smaller final λ (e.g., 0.8) to relax the worst-case requirement and reduce conservatism. This adjustment yields a non-empty polytope that no longer guarantees feasibility for every possible realization, but can still ensure decomposability with low error across the uncertainty distribution.

V. Case Studies

This section validates the proposed method using two aggregation case studies. The first case considers only continuous resource models, where existing methods [3], [4], [11], [13] are applicable. Comparisons show that the proposed method yields a tighter polytope fit than conventional aggregation models. The second case involves mixed-integer resource models and uncertainty. The results demonstrate that the proposed method maintains good accuracy and can yield a robust polytope. All code and data used in the case studies are made openly available for reproducibility in [34].

In all cases, the optimization horizon is set to 24 h with $T = 24$ discrete time intervals. All simulations are conducted on a desktop equipped with a 20-core Intel i7-14700KF CPU, an NVIDIA GeForce RTX 4070 GPU, and 32 GB of RAM. The proposed method and the approaches in [4], [11] are implemented in Python, while the methods in [3], [13] are implemented in MATLAB. In the polytope optimization procedure, directional error evaluation (Step 4) is solved on the CPU using Gurobi, whereas gradient computation and parameter updates (Steps 5-6) are performed on the GPU via PyTorch. The SGD algorithm is used for parameter updates.

A. Aggregating Deterministic Continuous Resource Models

This case studies the aggregation of continuous resource models, including TCLs, PVs, and CLs, aiming to verify the convergence behavior of the proposed method and compare its performance with existing approaches in [3], [4], [11], [13]. Although these resources are inherently subject to uncertainty, all uncertain parameters are fixed at their expected values in this case for simplicity, as this does not affect the objective of this test.

Two scenarios are considered. The first involves a small-scale aggregation with 50 resources, including 40 TCLs (heat pumps, indoor temperature 18-22°C, thermal parameters derived from realistic building data in Zurich, Switzerland), 5 PVs (30 kW installed capacity each, driven by real irradiance profiles with additional $\pm 20\%$ random variation across units to reflect device heterogeneity), and 5 CLs (30 kW baseline, driven by real load data with additional $\pm 20\%$ random variation across units, curtailment ratio 0.4). The second scenario considers a large-scale aggregation with 500 resources, including 200 TCLs, 200 PVs, and 100 CLs.

In the first scenario, all four benchmark methods in [3], [4], [11], [13] are evaluated, and the proposed method is initialized using the results from [11] and [4], respectively. Specifically, [11] provides an outer approximation defined by the PEB model ($M = 4T$), while [4] yields an inner approximation based on the PB model ($M = 2T$). The methods in [3] and [13] both generate inner approximations defined by the PEB model ($M = 4T$).

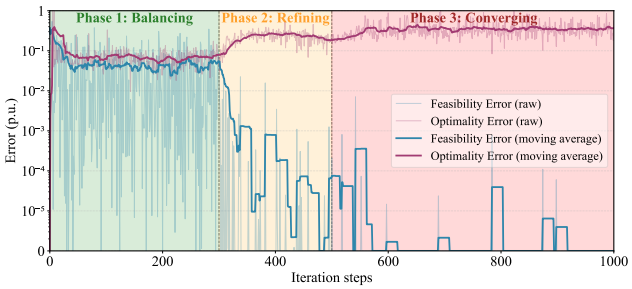
1) Iteration Process:

Under both initialization settings, the dynamic weighting strategy described in Subsection IV-E is adopted. In particular, a three-phase weighting scheme, as summarized in Table I, is applied to guide the polytope toward a tight inner approximation of the true feasible region.

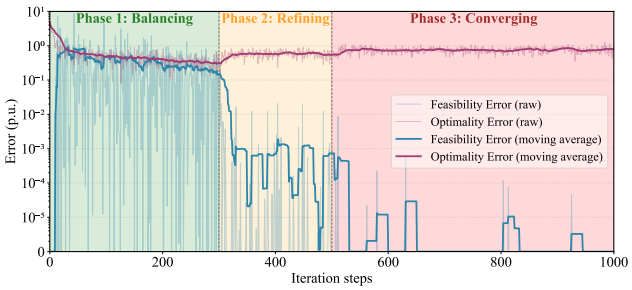
TABLE I: Optimization phases for the continuous case.

Phase	**Iterations**	λ	Learning rate for SGD
Phase 1: balancing	1–300	0.5	5×10^{-2}
Phase 2: refining	301–500	0.99	2×10^{-1}
Phase 3: converging	501–1000	$1 - 10^{-6}$	5×10^{-1}

Fig. 4 shows the evolution of feasibility and optimality errors over the three-phase optimization under both initialization settings. The errors are reported in per-unit values, as the aggregated power profile is normalized by the maximum adjustable range at each time interval. As shown in Fig. 4(a),



(a) PEB initialization ($M = 4T$)



(b) PB initialization ($M = 2T$)

Fig. 4: Evolution of feasibility and optimality errors during the training process with different initializations in the small-scale continuous case.

when initialized using [11], the feasibility error is initially large while the optimality error is zero, indicating that the initial polytope is an outer approximation. In Phase 1, both errors decrease to the level of 10^{-2}, suggesting that the polytope gradually captures the shape of the feasible region. In Phase 2, as λ increases to 0.99, feasibility is prioritized over optimality, leading to a significant reduction in feasibility error at the expense of increased optimality error. In Phase 3, with $\lambda = 1 - 10^{-6}$, the feasibility error is further reduced to a negligible level, and the resulting polytope can be regarded as an empirical inner approximation, while the optimality error increases as a trade-off. These results confirm that the proposed three-phase strategy functions as intended: Phase 1 identifies the polytope geometry, and Phases 2-3 refine it into a tight inner approximation. Fig. 4(b) shows the results when initialized using [4]. In this case, the feasibility error is initially zero, and the optimality error is high, confirming that the starting polytope is an inner approximation. After Phase 1, both errors reach a balanced level, but remain higher than those in Fig. 4(a). This is expected because the PB template has only half the number of facets compared to the PEB template, leading to lower approximation accuracy. This difference persists at convergence, where the final optimality error remains higher than that obtained in Fig. 4(a).

2) Comparative Analysis:

To further illustrate the accuracy advantage of the proposed method, the distributions of directional feasibility and optimality errors are evaluated for the final polytopes obtained by different methods. The statistics are computed over 100 sampled directions, as shown in Fig. 5. Note that the method in [3] is not included, as it fails to converge within 5000 s due

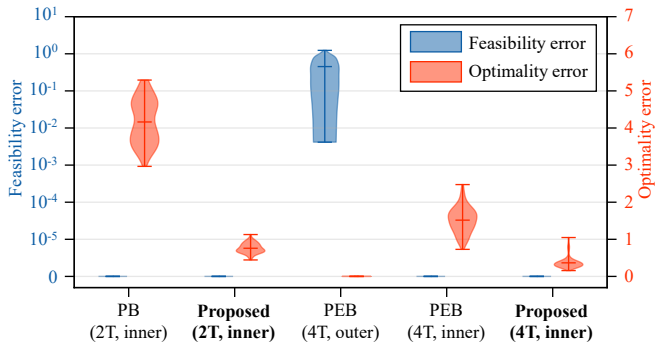
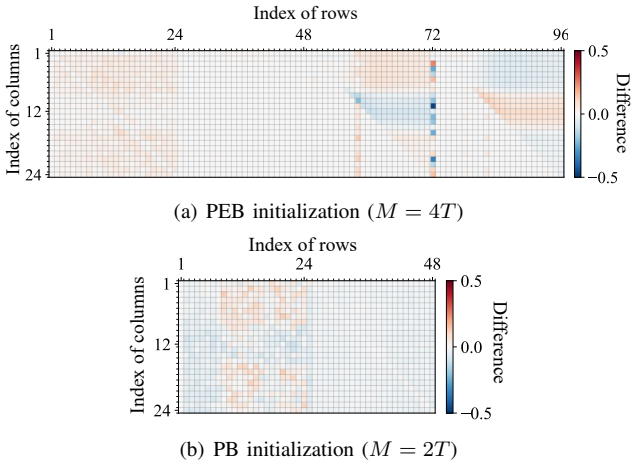


Fig. 5: Comparison of the proposed method with existing approaches. PB (2T, inner): [4]; PEB (4T, outer): [11]; PEB (4T, inner): [13]; Proposed (2T, inner): proposed method initialized with [4]; Proposed (4T, inner): proposed method initialized with [11].



(a) PEB initialization ($M = 4T$)

(b) PB initialization ($M = 2T$)

Fig. 6: Change in the polytope matrix $\mathbf{A}$ from initialization to convergence under different initializations in the small-scale continuous case (transposed).

to its high computational complexity. At the same complexity of $M = 2T$, the proposed method achieves a negligible feasibility error comparable to that of [4], while reducing the average optimality error by 81.7%. At $M = 4T$, the proposed method still reduces the feasibility error to a negligible level compared to [11], at the cost of only a marginal increase in optimality error. Compared with [13], which also maintains negligible feasibility error, the proposed method lowers the average optimality error by 75.9%. Moreover, comparing the two initialization settings of the proposed method reveals that the optimality error under $M = 4T$ is 51.8% lower than that under $M = 2T$ on average, confirming that a larger number of hyperplanes indeed yields a more accurate polytopic approximation.

Fig. 6 illustrates the change in the polytope matrix $\mathbf{A}$ from initialization to convergence under both initialization settings (shown in transposed form for compactness). The proposed method modifies $\mathbf{A}$ within the ranges of [-0.462, 0.243] and [-0.097, 0.121] under the PEB and PB initializations, respectively, which is the primary source of the accuracy improvements demonstrated above. Notably, the PEB template allows the proposed method to adjust both energy and power bounds, whereas the PB template lacks energy-bound parameters, restricting all modifications to power bounds alone. This greater flexibility in template structure explains why the PEB initialization achieves a lower optimality error than the PB initialization.

These results highlight a core advantage of the proposed method: rather than being confined to empirically designed polytope templates, it discovers polytope structures with significantly higher approximation accuracy for the aggregation task.

To further assess the practical impact of approximation accuracy, we conduct a cost optimization benchmark using the aggregated polytope models. Specifically, a linear program $\min_{\mathbf{P}} \mathbf{c}^\top \mathbf{P}$ subject to $\mathbf{AP} \leq \mathbf{b}$ is solved, where $\mathbf{c} \in \mathbb{R}^T$ is a cost coefficient vector. The matrices $\mathbf{A}$ and vectors $\mathbf{b}$ are obtained from each inner-approximation method, and $\mathbf{c}$ is sampled from a uniform distribution on $[0, 1]^T$ (CNY/kWh)

TABLE II: Cost optimization results under randomly generated cost vectors (100 trials).

Method	Mean Cost (CNY)	Cost Std (CNY)
PB ($2T$, inner) [4]	2116.66	264.90
PEB ($4T$, inner) [13]	1587.31	230.08
Proposed ($2T$, inner)	1470.03	216.82
Proposed ($4T$, inner)	1537.34	223.56

over 100 independent trials. Table II reports the mean and standard deviation of the resulting costs. The proposed method achieves lower average costs under both settings compared to the inner-approximation methods in [4] and [13], which directly reflects its reduced optimality error. An interesting observation is that the proposed method with the $2T$ template yields a lower average cost than with the $4T$ template, despite the latter having lower optimality error as shown in Fig. 5. This can be explained by the mismatch between training and evaluation distributions: the direction vectors $\mathbf{v}$ used in training are sampled from a standard normal distribution (symmetric around zero), whereas the cost vectors $\mathbf{c}$ in this benchmark are strictly positive. This observation suggests a potential improvement: when prior knowledge of the application (e.g., non-negative electricity prices) is available, the sampling distribution of $\mathbf{v}$ can be adapted accordingly instead of using a standard normal distribution, which may further enhance performance in practice.

3) Scalability Test:

We then evaluate the scalability of the proposed method under the second scenario of aggregating 500 resources. The same optimization strategy as in Table I is adopted, with initialization based on the outer approximation from the PEB model [11]. The algorithm completes 1000 iterations in 782.45 s, with an average of 0.78 s per iteration, demonstrating high computational efficiency. At convergence, the feasibility error is still negligible (below 10^{-5}), and the optimality error is 0.072. In comparison, the method in [4] also achieves negligible feasibility error but yields an optimality error of

TABLE III: Optimization phases for the mixed-integer case.

Phase	**Iterations**	λ	Learning rate for SGD
Phase 1: balancing	1–600	0.5	5×10^{-2}
Phase 2: refining	601–1200	0.99	1×10^{-1}
Phase 3: converging	1201–2000	$1 - 10^{-6}$	2×10^{-1}

4.32, which is about 60× higher. The methods in [3] and [13] encounter memory overflow and fail to complete the computation. These results indicate that the proposed method maintains high computational efficiency and consistently achieves significantly improved accuracy in large-scale aggregation tasks.

B. Aggregating Mixed-Integer Resource Models under Uncertainty

In this subsection, we leverage the proposed method to compute the aggregated flexibility region of resources with mixed-integer operational constraints. The study is based on a typical microgrid, consisting of one MT (20-100 kW, minimum on-time 4 h), one PV unit (200 kW installed capacity, driven by real irradiance profiles), one TL (70 kW, 280 kWh, 8:00-18:00), one CL (maximum 300 kW, curtailment ratio 0.4), two TCLs (heat pumps, indoor temperature 18-22 °C, rated power 185 kW, COP 3.85, thermal capacitance 4.61 kWh/°C, thermal conductance 372 kW/°C), and one ESS (250 kWh, 100 kW charge/discharge limits, charging/discharging efficiency 95%/98%, state-of-charge 10%-100%, initial state-of-charge 50%). The available PV power, CL, and ambient temperature profiles are varied within ±5% of their baseline values to create a dataset of uncertain parameter realizations. Additionally, the power adjustment ranges for PVs and CLs, as well as the temperature adjustment range for TCL, are reduced by 2% to enhance out-of-sample robustness. This case involves both integer variables (from MT, TL, and ESS) and uncertainty (from PV, CL, TCL), rendering the aggregation problem much more challenging than the continuous case. In fact, to the best of our knowledge, no existing aggregation method in the literature can handle such a mixed-integer scenario. Therefore, only the proposed method is implemented in this case to evaluate its general applicability.

We adopt a similar three-phase dynamic weighting strategy as in Subsection V-A, with adjusted parameters given in Table III. The number of training iterations is doubled to better handle uncertainty. Fig. 7 shows the evolution of feasibility and optimality errors. The overall trend is consistent with the continuous case, but with three notable differences. First, due to uncertainty, the initial polytope is not a strict outer approximation, leading to a nonzero initial feasibility error. Second, both the magnitude and fluctuations of the errors are larger than in the continuous case (Fig. 4(a)), reflecting the impact of uncertain variations. Third, despite these challenges, the feasibility error is eventually reduced to a negligible level at the end of Phase 3, indicating that the final polytope $\mathcal{P}(\mathbf{A}^*, \mathbf{b}^*)$ provides a numerically feasible inner approximation under uncertainty.

To further validate the robustness of the obtained polytope model, we conduct an out-of-sample test using 100 additional

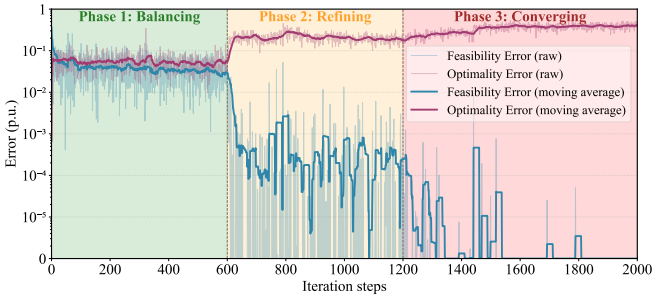


Fig. 7: Evolution of feasibility and optimality errors during the training process for the mixed-integer case.

uncertain parameter scenarios not in the original dataset. For each test scenario, we randomly sample an aggregated power profile from the polytope and try to decompose it into feasible individual resource schedules by minimizing the disaggregation error (see [12]). Notably, all 100 test scenarios achieve numerically zero disaggregation error (maximum 1.02×10^{-6}), revealing the robustness of the polytopic aggregation against various uncertainty realizations.

These results demonstrate that the proposed method is sufficiently general to handle mixed-integer resources under uncertainty while maintaining high solution quality.

The entire procedure completes in 225.89 s, with an average of 0.11 s per iteration. The relatively low computational cost is due to the small number of resources in this case. For larger systems with many mixed-integer components, the computational burden is expected to increase, mainly due to the intrinsic complexity of mixed-integer optimization rather than the proposed method itself. Advances in mixed-integer solvers are expected to further improve the efficiency of the overall iteration process.

VI. Conclusions

This paper proposes a polytope fitting method to characterize the aggregated flexibility region of heterogeneous distributed resources. By jointly optimizing the polytope geometry matrix and boundary vector via approximate gradient descent, the proposed method significantly improves representation accuracy compared to existing empirically designed aggregation models. The results demonstrate that the method can effectively handle large-scale aggregation, mixed-integer resource models, and uncertainty, while maintaining high accuracy across different cases, indicating its strong generality.

Several directions remain for future improvement. These include establishing a more rigorous theoretical analysis of the gradient approximations, developing more effective dynamic weighting strategies to accelerate convergence, and extending the framework to handle nonconvex settings. In addition, future work may also incorporate cost functions into the aggregation process via epigraph representations and explore the applicability of the proposed framework to a broader range of domains.

APPENDIX A
INDIVIDUAL OPERATIONAL MODELS OF REPRESENTATIVE FLEXIBLE RESOURCES

This appendix presents the operational constraints of six representative flexible resources: microturbines (MTs), PVs, ESSs, curtailable loads (CLs), transferable loads (TLs), and TCLs. The proposed aggregation framework is sufficiently general and can be readily extended to accommodate a wide range of other resource types.

1) Microturbines: Microturbines are controllable distributed generation units that can provide flexibility to the power system. The operational constraints are:

$$p^{\text{MT}}_{\text{min},i} u^{\text{MT}}_{i,t} \le p_{i,t} \le p^{\text{MT}}_{\text{max},i} u^{\text{MT}}_{i,t}, \forall t \in [T] \tag{A.1a}$$

$$u^{\text{MT}}_{i,t} - u^{\text{MT}}_{i,t-1} = y^{\text{MT}}_{i,t} - z^{\text{MT}}_{i,t}, \forall t \in [T] \tag{A.1b}$$

$$y^{\text{MT}}_{i,t} + z^{\text{MT}}_{i,t} \le 1, \forall t \in [T] \tag{A.1c}$$

$$u^{\text{MT}}_{i,t} \ge \sum_{\tau=t}^{t+t^{\text{MT}}_{\text{on},i}-1} y^{\text{MT}}_{i,\tau}, \forall t \in [T - t^{\text{MT}}_{\text{on},i} + 1] \tag{A.1d}$$

where $p^{\text{MT}}_{\text{min},i}$ and $p^{\text{MT}}_{\text{max},i}$ are the MT's minimum and maximum power limits, and $t^{\text{MT}}_{\text{on},i}$ is the minimum on-time requirement. The three binary variables $u^{\text{MT}}_{i,t} \in \{0,1\}$ (commitment status), $y^{\text{MT}}_{i,t} \in \{0,1\}$ (startup action), and $z^{\text{MT}}_{i,t} \in \{0,1\}$ (shutdown action) at each time $t \in [T]$ are necessary for formulating the minimum generation output constraint (A.1a) and the minimum on-time requirement (A.1d). However, the presence of these binary variables introduces discontinuities in the operational region, thereby increasing the complexity of flexibility modeling.

2) Photovoltaics: PVs are characterized as fluctuating renewable energy sources with stochastic availability. They offer flexibility in that their power output is controllable within a range from zero to the available instantaneous capacity. For each PV i, the power output is constrained by:

$$0 \le p_{i,t} \le \tilde{p}^{\text{PV}}_{i,t}, \quad \forall t \in [T] \tag{A.2}$$

where $\tilde{p}^{\text{PV}}_{i,t}$ denotes the available (maximum) power of PV i at time t determined by solar irradiance and panel efficiency. The parameter $\tilde{p}^{\text{PV}}_{i,t}$ is uncertain due to unpredictable weather conditions, which introduces variability in the flexibility region boundary across different realizations.

3) Transferable Loads: TLs represent deferrable energy consumption tasks, such as electric vehicle (EV) charging and industrial processes. They provide flexibility by allowing the energy consumption to be shifted across time within a specific operating window, thus enabling the system to adjust the load profile according to operational needs. The operational constraints are:

$$-p_{i,t} = p^{\text{TL}}_{\text{rated},i} u^{\text{TL}}_{i,t}, \forall t \in [T] \tag{A.3a}$$

$$u^{\text{TL}}_{i,t} = 0, \forall t \notin [t^{\text{TL}}_{\text{start},i}, t^{\text{TL}}_{\text{end},i}) \tag{A.3b}$$

$$-\sum_{t\in[T]} p_{i,t}\Delta t \ge e^{\text{TL}}_{\text{req},i} \tag{A.3c}$$

where $p^{\text{TL}}_{\text{rated},i}$ is the rated power (consumption is represented as negative power), $[t^{\text{TL}}_{\text{start},i}, t^{\text{TL}}_{\text{end},i})$ denotes the allowable operation time window, and $e^{\text{TL}}_{\text{req},i}$ is the total energy requirement. Constraint (A.3a) couples the power with the binary status $u^{\text{TL}}_{i,t} \in \{0,1\}$; (A.3b) restricts operation to the specified window; and (A.3c) ensures the total energy consumption task is completed. Similar to MTs, the binary variable $u^{\text{TL}}_{i,t}$ introduces discontinuity in the flexibility region by creating discrete operating modes. It is worth noting that the above modeling of TLs is general: the corresponding entity may represent an entire EV charging station or a factory with many devices. Should detailed data be available, such as individual EV charging orders or specific device schedules, refined modeling approaches, such as the EV charging flexibility model [35] and specific device operational models, could be adopted. Crucially, the application of more detailed models does not compromise the validity of the proposed method.

4) Curtailable Loads: CLs correspond to demand response programs where consumers agree to reduce consumption in exchange for compensation. They offer continuous flexibility because their power consumption can be adjusted within a specific range. For each CL i, the power consumption (negative power) is constrained by:

$$(1-\rho^{\text{CL}}_i)\tilde{p}^{\text{CL}}_{i,t} \le -p_{i,t} \le \tilde{p}^{\text{CL}}_{i,t}, \quad \forall t \in [T] \tag{A.4}$$

where $\tilde{p}^{\text{CL}}_{i,t}$ is the baseline (maximum) load profile, and $\rho^{\text{CL}}_i \in [0,1]$ is the maximum curtailment ratio. The baseline load profile $\tilde{p}^{\text{CL}}_{i,t}$ is uncertain due to unpredictable consumer behavior, which introduces variability in the boundary of the flexibility region across different scenarios.

5) Energy Storage Systems: ESSs provide temporal energy shifting capabilities through charging and discharging operations. They are able to provide flexibility by decoupling the timing of energy production and consumption, effectively acting as a buffer to balance supply and demand. For each ESS i, let $e^{\text{ESS}}_{i,t}$ denote the stored energy at time t, and introduce binary variable $u^{\text{ESS}}_{i,t} \in \{0,1\}$ to prevent simultaneous charging and discharging. The operational constraints are:

$$e^{\text{ESS}}_{i,t} = e^{\text{ESS}}_{i,t-1} + \Delta t \left(p^{\text{ESS}}_{\text{chg},i,t}\eta^{\text{ESS}}_{\text{chg}} - \frac{p^{\text{ESS}}_{\text{dis},i,t}}{\eta^{\text{ESS}}_{\text{dis}}} \right), \forall t \in [T] \tag{A.5a}$$

$$e^{\text{ESS}}_{\text{min},i} \le e^{\text{ESS}}_{i,t} \le e^{\text{ESS}}_{\text{max},i}, \forall t \in [T] \tag{A.5b}$$

$$0 \le p^{\text{ESS}}_{\text{chg},i,t} \le p^{\text{ESS}}_{\text{chg},i} u^{\text{ESS}}_{i,t}, \forall t \in [T] \tag{A.5c}$$

$$0 \le p^{\text{ESS}}_{\text{dis},i,t} \le p^{\text{ESS}}_{\text{dis},i}(1-u^{\text{ESS}}_{i,t}), \forall t \in [T] \tag{A.5d}$$

$$p_{i,t} = p^{\text{ESS}}_{\text{dis},i,t} - p^{\text{ESS}}_{\text{chg},i,t}, \forall t \in [T] \tag{A.5e}$$

$$e^{\text{ESS}}_{i,T} \ge e^{\text{ESS}}_{i,0} = e^{\text{ESS}}_{\text{init},i} \tag{A.5f}$$

where $p^{\text{ESS}}_{\text{chg},i,t}$ and $p^{\text{ESS}}_{\text{dis},i,t}$ are the charging and discharging power at time t; $\eta^{\text{ESS}}_{\text{chg}}$ and $\eta^{\text{ESS}}_{\text{dis}}$ are charging and discharging efficiencies; $p^{\text{ESS}}_{\text{chg},i}$ and $p^{\text{ESS}}_{\text{dis},i}$ are charging and discharging power limits; and $e^{\text{ESS}}_{\text{min},i}$, $e^{\text{ESS}}_{\text{max},i}$, and $e^{\text{ESS}}_{\text{init},i}$ are the minimum, maximum, and initial energy. Constraint (A.5a) describes the power-energy dynamics; (A.5b) enforces energy capacities; (A.5c) and (A.5d) impose charging and discharging power limits with mutual exclusivity; and (A.5e) defines the net power injection/withdrawal.

6) Thermostatically Controlled Loads: TCLs, such as heat pumps and air conditioners, provide space heating and cooling

with inherent thermal inertia. They represent a source of flexibility by utilizing the building's thermal mass to store energy, allowing power consumption to be adjusted over short periods without significantly compromising user comfort. For each TCL i, let variable $\theta^{\text{in}}_{i,t}$ denote the indoor temperature at time t. The operational constraints are:

$$\theta^{\text{in}}_{i,t} = \alpha^{\text{TCL}}_i \theta^{\text{in}}_{i,t-1} + (1-\alpha^{\text{TCL}}_i)\left(\tilde{\theta}^{\text{amb}}_t \mp \frac{\eta^{\text{TCL}}_i}{h^{\text{TCL}}_i} p_{i,t}\right), \forall t \in [T] \tag{A.6a}$$

$$\theta^{\text{TCL}}_{\min,i} \le \theta^{\text{in}}_{i,t} \le \theta^{\text{TCL}}_{\max,i}, \forall t \in [T] \tag{A.6b}$$

$$0 \le -p_{i,t} \le p^{\text{TCL}}_{\max,i}, \forall t \in [T] \tag{A.6c}$$

where $\tilde{\theta}^{\text{amb}}_t$ is the ambient temperature; h^{TCL}_i and c^{TCL}_i are thermal conductance and capacitance; η^{TCL}_i is the coefficient of performance (COP); $p^{\text{TCL}}_{\max,i}$ is the maximum output power; the sign $\mp$ takes "$-$" for heating TCLs and "$+$" for cooling TCLs; $\alpha^{\text{TCL}}_i = \exp(-\Delta t \cdot h^{\text{TCL}}_i / c^{\text{TCL}}_i)$ is the thermal decay factor; and $\theta^{\text{TCL}}_{\min,i}$, $\theta^{\text{TCL}}_{\max,i}$ are comfort temperature bounds. The initial temperature satisfies $\theta^{\text{in}}_{i,0} = \theta^{\text{TCL}}_{\text{set},i}$. The ambient temperature $\tilde{\theta}^{\text{amb}}_t$ is subject to uncertainty due to weather variations, which affects the temperature dynamics and consequently the flexibility region.

REFERENCES

[1] Y. Fan, X. Wu, B. Cao, and X. Wang, "Multi-stage robust feasible region based integrated energy microgrid dispatch," *CSEE Journal of Power and Energy Systems*, pp. 1–12, 2024, early access.

[2] Y. Lin, Z. Lu, Y. Qiao, and R. Chen, "Flexible region aggregation of adjustable loads via an adaptive convex hull strategy," *Global Energy Interconnection*, vol. 8, no. 1, pp. 106–120, 2025.

[3] S. Wang and W. Wu, "Aggregate flexibility of virtual power plants with temporal coupling constraints," *IEEE Transactions on Smart Grid*, vol. 12, no. 6, pp. 5043–5051, 2021.

[4] X. Chen, E. Dall'Anese, C. Zhao, and N. Li, "Aggregate power flexibility in unbalanced distribution systems," *IEEE Transactions on Smart Grid*, vol. 11, no. 1, pp. 258–269, Jan. 2020.

[5] G. Tsaousoglou, I. Sartzetakis, P. Makris, N. Efthymiopoulos, E. Varvarigos, and N. G. Paterakis, "Flexibility aggregation of temporally coupled resources in real-time balancing markets using machine learning," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 7, pp. 4342–4351, 2022.

[6] X. Wang, J. Ke, H. Wu, Y. Dong, C. Jiang, W. Huang, and S. Zhang, "A robust safe reinforcement learning approach for power grid resilience enhancement against typhoons via der flexibility aggregation," *IEEE Transactions on Industry Applications*, pp. 1–12, 2025.

[7] K. Zhang, X. Wang, M. Shahidehpour, C. Jiang, H. Yang, Z. Ding, and Z. Li, "Inverse optimization and robust aggregation based bidding strategy for distributed energy resource aggregators using multi-agent reinforcement learning," *International Journal of Electrical Power & Energy Systems*, vol. 174, p. 111468, 2026.

[8] Z. Tan, Z. Yan, Q. Xia, and Y. Wang, "Data-driven inverse optimization for modeling intertemporally responsive loads," *IEEE Transactions on Smart Grid*, vol. 14, no. 5, pp. 4129–4132, 2023.

[9] R. Lyu, H. Guo, G. Strbac, and C. Kang, "Data-driven dimension reduction for industrial load modeling using inverse optimization," *IEEE Transactions on Smart Grid*, vol. 16, no. 3, pp. 2695–2698, 2025.

[10] S. Taheri, V. Kekatos, S. Veeramachaneni, and B. Zhang, "Data-driven modeling of aggregate flexibility under uncertain and non-convex device models," *IEEE Transactions on Smart Grid*, vol. 13, no. 6, pp. 4572–4582, 2022.

[11] Z. Xu, D. S. Callaway, Z. Hu, and Y. Song, "Hierarchical coordination of heterogeneous flexible loads," *IEEE Transactions on Power Systems*, vol. 31, no. 6, pp. 4206–4216, Nov. 2016.

[12] Y. Wen, Z. Hu, S. You, and X. Duan, "Aggregate feasible region of DERs: Exact formulation and approximate models," *IEEE Transactions on Smart Grid*, vol. 13, no. 6, pp. 4405–4423, 2022.

[13] Y. Wen, Z. Hu, J. He, and Y. Guo, "Improved inner approximation for aggregating power flexibility in active distribution networks and its applications," *IEEE Transactions on Smart Grid*, vol. 15, no. 4, pp. 3653–3665, 2024.

[14] Y. Zhou, C. Essayeh, and T. Morstyn, "Aggregated feasible active power region for distributed energy resources with a distributionally robust joint probabilistic guarantee," *IEEE Transactions on Power Systems*, vol. 40, no. 1, pp. 556–571, 2025.

[15] S. Wang, W. Wu, Q. Chen, J. Yu, and P. Wang, "Stochastic flexibility evaluation for virtual power plants by aggregating distributed energy resources," *CSEE Journal of Power and Energy Systems*, vol. 10, no. 3, pp. 988–999, 2024.

[16] X. Chen and N. Li, "Leveraging two-stage adaptive robust optimization for power flexibility aggregation," *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 3954–3965, 2021.

[17] L. Yuan, Q. Yu, X. Yao, and M. Guo, "Aggregation feasible region of full power spectrum electric vehicles based on polyhedron projection," *IEEE Transactions on Sustainable Energy*, vol. 16, no. 3, pp. 2029–2043, 2025.

[18] M. Zhang, Y. Xu, X. Shi, and Q. Guo, "A fast polytope-based approach for aggregating large-scale electric vehicles in the joint market under uncertainty," *IEEE Transactions on Smart Grid*, vol. 15, no. 1, pp. 701–713, 2024.

[19] L. Zhao, W. Zhang, H. Hao, and K. Kalsi, "A geometric approach to aggregate flexibility modeling of thermostatically controlled loads," *IEEE Transactions on Power Systems*, vol. 32, no. 6, pp. 4721–4731, 2017.

[20] S. Córdova, C. A. Cañizares, A. Lorca, and D. E. Olivares, "Aggregate modeling of thermostatically controlled loads for microgrid energy management systems," *IEEE Transactions on Smart Grid*, vol. 14, no. 6, pp. 4169–4181, 2023.

[21] Y. Wen, B. Zhao, Y. Guo, and Z. Hu, "Equivalent storage homogenization of thermostatically controlled loads via dynamic programming," *IEEE Transactions on Smart Grid*, vol. 16, no. 6, pp. 5693–5696, 2025.

[22] M. S. Nazir, I. A. Hiskens, A. Bernstein, and E. Dall'Anese, "Inner approximation of minkowski sums: A union-based approach and applications to aggregated energy resources," in *2018 IEEE Conference on Decision and Control (CDC)*, 2018, pp. 5708–5715.

[23] Z. Yi, Y. Xu, W. Gu, L. Yang, and H. Sun, "Aggregate operation model for numerous small-capacity distributed energy resources considering uncertainty," *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 4208–4224, 2021.

[24] T. Zhang, J. Wang, G. Li, X. Wang, and M. Zhou, "Characterizing temporal-coupled feasible region of active distribution networks," *IEEE Transactions on Industry Applications*, vol. 58, no. 5, pp. 5687–5696, 2022.

[25] Z. Tan, H. Zhong, J. Wang, Q. Xia, and C. Kang, "Enforcing intra-regional constraints in tie-line scheduling: A projection-based framework," *IEEE Transactions on Power Systems*, vol. 34, no. 6, pp. 4751–4761, 2019.

[26] R. M. Starr, *Shapley–Folkman Theorem*. London: Palgrave Macmillan UK, 2017, pp. 1–3.

[27] P. Jungeblut, L. Kleist, and T. Miltzow, "The complexity of the Hausdorff distance," *Discrete & Computational Geometry*, vol. 71, no. 1, pp. 177–213, Jan. 2024.

[28] Y. Bengio, N. Léonard, and A. Courville, "Estimating or propagating gradients through stochastic neurons for conditional computation," 2013. [Online]. Available: https://arxiv.org/abs/1308.3432

[29] A. Paszke, S. Gross, F. Massa *et al.*, "Pytorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32. Vancouver, Canada: Curran Associates, Inc., 2019.

[30] M. Abadi, P. Barham, J. Chen *et al.*, "Tensorflow: A system for large-scale machine learning," in *12th USENIX Symposium on Operating Systems Design and Implementation (OSDI 16)*, 2016, pp. 265–283.

[31] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, no. 6088, pp. 533–536, Oct. 1986.

[32] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *3rd International Conference on Learning Representations, ICLR 2015, Conference Track Proceedings*, San Diego, USA, 2015.

[33] B. Stott and O. Alsac, "Fast decoupled load flow," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-93, no. 3, pp. 859–869, 1974.

[34] Y. Wen, "PolytopeAggregation," 2024. [Online]. Available: https://github.com/wenyl16/PolytopeAggregation

[35] H. Zhang, Z. Hu, Z. Xu, and Y. Song, "Evaluation of achievable vehicle-to-grid capacity using aggregate pev model," *IEEE Transactions on Power Systems*, vol. 32, no. 1, pp. 784–794, 2017.